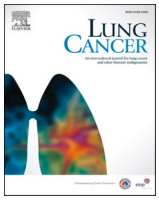
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Revisiting the lung cancer screening eligibility criteria to promote equity for Black individuals

Abdi T. Gudina [a,*], Charles Kamen [b], Sara J. Hardy [c,d], Lee Kehoe [b], Eva Culakova [b], Ana-Paula Cupertino [e]

[a] Department of Public Health Sciences at the University of Rochester, School of Medicine and Dentistry, NY, United States
[b] Division of Supportive Care in Cancer in the Department of Surgery at the University of Rochester, School of Medicine and Dentistry and the James P. Wilmot Cancer Institute at the University of Rochester, School of Medicine and Dentistry, NY, United States
[c] Department of Radiation Oncology, University of Rochester Medical Center, Rochester, NY, United States
[d] Department of Neurology, University of Rochester Medical Center, Rochester, NY, United States
[e] Surgical Health Outcomes and Reaching for Equity (SHORE), Department of Surgery, University of Rochester Medical Center, Rochester, NY, United States

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ABSTRACT

Background: Early detection using low-dose computed tomography reduces lung-cancer-specific mortality by 20% among high-risk individuals. Blacks are less likely than Whites to meet lung cancer screening (LCS) criteria under both the former and the updated United States Preventive Services Task Force (USPSTF) guidelines. The purpose of this study was to assess racial disparities in LCS eligibility and to propose tailored eligibility criteria for Blacks to enable equitable screening rate between Whites and Blacks.

Methods: Data for this study were obtained from the Behavioral Risk Factor Surveillance System (2017–2021). 101,552 subjects were included in the final analysis. By employing a systematic approach, we sought cut-off points at which Blacks were equally likely as Whites to be eligible for LCS. We evaluated the minimum age and smoking pack-years for Blacks while we retained the 2021 USPSTF criteria for Whites. The final decision was based on the minimum Wald's Chi-square statistics.

Results: The model we employed identified cut-off points at which Blacks were equally likely as Whites to be eligible for LCS. Retaining the 2021 USPSTF criteria for Whites, the model discovered a new pair of points for Blacks by reducing the minimum age to 43 years and decreasing the cumulative number of cigarettes smoked to 15 pack-years. Based on these cut-off points, we created tailored criteria for Blacks. Under the tailored criteria, Blacks (OR: 1.00; 95 %CI: 0.88–1.14) had the same odds of eligibility for LCS as Whites. The odds of eligibility for LCS by sex under the tailored criteria did not differ significantly for Black men (OR: 1.02; 95 %CI: 0.85–1.24) and Black women (OR: 0.95; 95 %CI: 0.81–1.12) compared to their respective White counterparts.

Conclusions: These tailored criteria for Blacks eliminate the disparities between Blacks and Whites in LCS eligibility. Future studies should test the sensitivity and specificity of these tailored criteria.

1. Introduction

Lung cancer is the leading cause of cancer death in the United States (U.S) in men and women accounting for about 25 % of all cancer mortality [1–3]. Black individuals are disproportionately affected by lung cancer compared with all other racial/ethnic groups in the U.S [4,5]. Black individuals smoke fewer cigarettes [6,7] and start smoking later in life [8,9], yet they are at greater risk of developing lung cancer than Whites [10] even after controlling for socioeconomic status and other risk factors [10–12]. Moreover, Black individuals are more likely to be diagnosed at younger ages [13] and with late-stage lung cancers than Whites [4].

Screening with low dose computerized tomography (LDCT) detects lung cancer at earlier, more treatable stages, thus improving clinical outcomes. In 2011, the U.S. National Lung Cancer Screening Trial (NLST) reported a 20 % reduction in lung cancer specific mortality with the use of LDCT scanning, compared to screening with the use of chest radiography [14]. In 2013, following the finding by the NLST, the

* Corresponding author at: University of Rochester Medical Center, Department of Public Health Sciences, 265 Crittenden Blvd, Rm 3-225, Rochester, NY 14642, United States.
E-mail address: Abdi_Gudina@urmc.rochester.edu (A.T. Gudina).

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United States Preventive Services Task Force (USPSTF) recommended annual screening for lung cancer using LDCT scanning for high-risk individuals (aged 55–80 years with a ≥30 pack-year smoking history and currently smoking or formerly smoked who quit smoking within the past 15 years) [15].

Under the 2013 USPSTF guideline, racial/ethnic minorities, particularly Blacks, were less likely to be eligible for lung cancer screening than Whites [11]. In 2021, the USPSTF updated its 2013 guideline for LCS eligibility by reducing the minimum age from 55 to 50 years and decreasing the cumulative number of cigarette pack-years smoked from 30 to 20, partly to mitigate the racial/ethnic disparities encountered under the 2013 eligibility criteria [16]. Although the updated guideline (i.e., the 2021 USPSTF) increased the number of eligible individuals for LCS, Black individuals remained less likely to be eligible for the screening [17,18]. Without tailored eligibility criteria, racial/ethnic minorities, particularly Blacks, may continue to be less likely to be eligible for LCS [17]. To our knowledge, no tailored criteria have been proposed or recommended for LCS in Blacks. The purpose of this study was to assess racial disparities (Blacks vs. Whites) in LCS eligibility and to propose tailored eligibility criteria for Blacks to enable equitable screening rates for Blacks and Whites.

2. Methods

2.1. Data source and study population

Data for this study were obtained from the 2017–2021 BRFSS, a population-based survey administered via cell phone and landline and conducted annually by the Centers for Disease Control and Prevention (CDC) in collaboration with health departments in all 50 states, Washington, DC, and the U.S territories. The BRFSS collects data on health conditions, preventive health practices, and risk behaviors of adults. The BRFSS methods (sample selection, including the weighting procedure and technical information) are described elsewhere [19]. All BRFSS questionnaires, data, and reports are available at http://www.cdc.gov/brfss/ [20]. Although the BRFSS collects data from all 50 states, Washington, DC, and the U.S territories, the optional module of the BRFSS covering lung cancer screening was administered in only 29 states from 2017 to 2021.

2.2. Selecting study participants from BRFSS

Our analyses were restricted to ever smoked individuals in the 29 U. S. states where the BRFSS survey was administered to collect data on LCS from 2017 to 2021. Sample selection for the final analysis is shown in the flow chart (see Fig. 1). Never smoked individuals, races/ethnicities other than Blacks or Whites, and those with missing information on the variables used to determine USPSTF eligibility criteria were excluded from the analysis. In addition, because the BRFSS data collapsed all people older than 80 years into the 80-year-old age group, we excluded this age group to avoid the inclusion of individuals older than 80 years who are ineligible for screening per USPSTF recommendation; this strategy may have excluded individuals aged exactly 80 years who would otherwise have been eligible. 101,552 subjects were included in the final analysis.

2.3. Outcome of interest

The outcome of interest for this study was eligibility (yes/no) for LCS per the USPSTF recommendation (using the 2013 or 2021 guidelines) or the newly created tailored criteria for Blacks. The criteria for LCS eligibility were determined by age, pack-year smoking history, and time since quit smoking. We used the average number of cigarettes smoked per day and the number of years of smoking to calculate pack-year

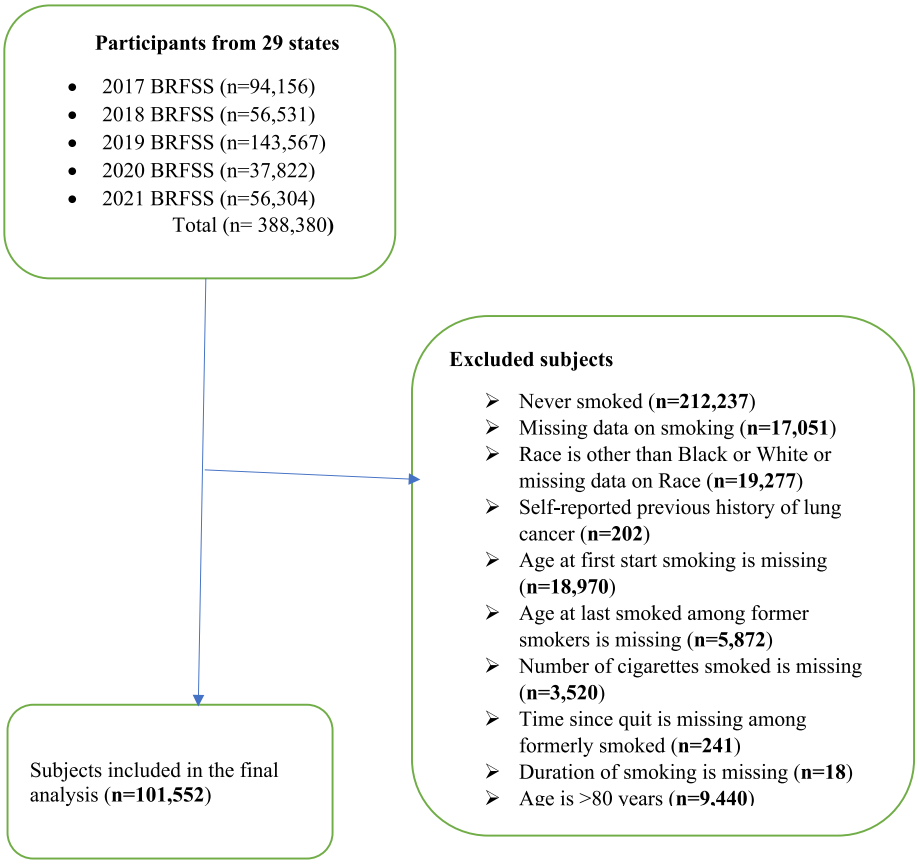


Fig. 1. Flow Chart of the study population and sample size.

smoking history.

2.4. Predictor variables.

The predictor of interest was racial groups which included Non-Hispanic White and Non-Hispanic Black.

2.5. Covariates of interest

Sociodemographic and health-related variables: income, marital status, education, insurance (health care coverage), COPD, survey year, and primary care provider (yes/no) were considered covariates of interest.

2.6. Statistical analysis

Frequency and percentage (for categorical variables) were used to describe the characteristics of the study population by race/ethnicity. We assessed racial disparities (Blacks vs. Whites) in LCS eligibility by the USPSTF guidelines (i.e., $USPSTF_{2013}$ and $USPSTF_{2021}$) using multivariable logistic regression models (in which survey weights, sampling cluster, and strata were considered) adjusting for potential confounders (income, marital status, education, insurance (health care coverage), COPD, primary care provider, and survey year).

We sought a cut-off point for age and smoking pack-years at which Blacks were equally likely to be eligible for LCS as Whites. To do so, we employed a systematic approach evaluating age and smoking pack-years at varying points. In Black individuals, for the age lower bound we considered options from 40 to 50 years, and for smoking pack-years we considered options from 10 to 20. Since we were tailoring for Blacks, we fixed the cut-off point for Whites according to the $USPSTF_{2021}$ guideline (i.e., aged 50–80 years, $\geq$20 pack-years, $\leq$15 years since quit smoking). For each pair of age and smoking pack-years, the algorithm first systematically evaluated the potential eligibility for LCS which resulted in 11 options for age and 11 options for smoking pack-years (11*11 = 121 options). Using the algorithm for each pair (age, smoking pack-years), the potential eligibility was recalculated and the logistic regression model was applied. From 121 options, we selected the option with the minimum Wald's Chi-square statistics resulting in odds ratio close to 1. The pair of age and smoking pack-years at which the odds ratio was close to 1 was found to be age 43 years and 15 pack-years. Utilizing these cut-off points (age 43 years and 15 pack-years) for the tailored criteria, we ran multivariable logistic regression models to reassess disparities between the racial/ethnic groups (Blacks vs Whites).

3. Results

3.1. Characteristics of the study population

Table 1 shows the characteristics of study populations by race/ethnicity. Of the 101,552 ever smoked adults included in the final analysis, the majority were White (93.32 %) (not shown in the table), aged 50 years and above (71.72 %), formerly smoked (64.77 %), had health insurance (91.39 %), and had a primary care provider (85.59 %). Among Whites, the majority of the participants were aged 50–89 years (71.77 %), had greater than $50,000.00 annual family income (56.01 %), had health insurance (91.59 %), had no COPD (83.54 %), formerly smoked (65.63 %), and had a primary care provider (85.65 %). Among Blacks, majority were aged 50–89 years (71.03 %), female gender (53.13 %), had health insurance (88.58 %), had no COPD (85.09 %), formerly smoked (52.66 %), and had a primary care provider (84.72 %).

3.2. Proportion of ever smoked individuals eligible for LCS under different guidelines

Under the $USPSTF_{2013}$ guideline, only 9.12 % of Blacks and 17.26 %

Table 1
Characteristics of the study population by Race/ethnicity (n = 101,552).

Characteristic	Race/Ethnicity, n (%)			P-value
	Overall (n = 101,552)	Whites, n = 94,765 (93.32%)	Blacks, n = 6,787 (6.68%)	
Age (in year)				0.1945
<50	28721 (28.28)	26755 (28.23)	1966 (28.97)	
50–79	72831 (71.72)	68010 (71.77)	4821 (71.03)	
Gender				**0.0015**
Male	49717 (48.96)	46538 (49.11)	3179 (46.84)	
Female	51808 (51.02)	48202 (50.86)	3606 (53.13)	
Refused/do not know/not sure	27 (0.03)	25 (0.03)	2 (0.03)	
Marital status				**<.0001**
Married couple	51708 (50.92)	49713 (52.46)	1995 (29.39)	
Divorced/ separated/ widowed	32377 (31.88)	29777 (31.42)	2600 (38.31)	
Never married	17089 (16.83)	14943 (15.77)	2146 (31.62)	
Refused/do not know/not sure	378 (0.37)	332 (0.35)	46 (0.68)	
Education				**<.0001**
Less than high school	1305 (1.29)	1189 (1.25)	116 (1.71)	
High school or equivalent	38176 (37.59)	35084 (37.02)	3092 (45.56)	
Some college	32221 (31.73)	30102 (31.76)	2119 (31.22)	
College graduate and above	29728 (29.27)	28278 (29.84)	1450 (21.36)	
Refused/do not know/not sure	122 (0.12)	112 (0.12)	10 (0.15)	
Income				**<.0001**
<$15,000.0	8385 (8.26)	7296 (7.70)	1089 (16.05)	
$15,000.00-$50,000.00	37258 (36.69)	34393 (36.29)	2865 (42.21)	
>$50,000.00	55909 (55.05)	53076 (56.01)	2833 (41.74)	
Insurance (Health Care Coverage)				**<.0001**
Yes	92808 (91.39)	86796 (91.59)	6012 (88.58)	
No	8207 (8.08)	7506 (7.92)	701 (10.33)	
Refused/do not know/not sure	537 (0.53)	463 (0.49)	74 (1.09)	
COPD				**0.0021**
Yes	16061 (15.82)	15089 (15.92)	972 (14.32)	
No	84939 (83.64)	79164 (83.54)	5775 (85.09)	
Refused/do not know/not sure	552 (0.54)	512 (0.54)	40 (0.59)	
Smoking history				**<.0001**
Formerly smoked	65772 (64.77)	62198 (65.63)	3574 (52.66)	
Currently smoking	35780 (35.23)	32567 (34.37)	3213 (47.34)	
Primary Care Provider				0.1045
Yes	86916 (85.59)	81166 (85.65)	5750 (84.72)	
No	14336 (14.12)	13319 (14.05)	1017 (14.98)	
Refused/do not know/not sure	300 (0.29)	280 (0.30)	20 (0.29)	

of Whites were eligible for LCS among ever smoked adults. Under the $USPSTF_{2021}$ guideline, which was developed by reducing the minimum age from 55 to 50 years and by decreasing the minimum smoking pack-years from 30 to 20 pack-years, the proportion of LCS eligible individuals was doubled among Blacks (from 9.12 % to 18.49 %) and was raised from 17.26 % to 26.60 % among Whites ever smoked adults. The tailored criteria increased the proportion of eligible individuals to 26.34 % among Blacks. Under all guidelines including the tailored criteria, more men were identified as eligible for LCS than women (see Table 2).

3.3. Disparities in lung cancer screening eligibility by race/ethnicity under different guidelines

Table 3 shows the odds of LCS eligibility among ever smoked adults comparing Whites and Blacks using the USPSTF guidelines (i.e., $USPSTF_{2013}$ and $USPSTF_{2021}$) and tailored criteria. Under the $USPSTF_{2013}$ guideline, Blacks (AOR: 0.46; 95 %CI: 0.38–0.54) were significantly less likely than Whites to be eligible for lung cancer screening. We further stratified the data by sex to evaluate if LCS eligibility differs by race and sex comparing Blacks with Whites by sex. We found that under the $USPSTF_{2013}$ guideline, Black men (AOR: 0.50; 95 %CI: 0.38–0.64) and Black women (AOR: 0.38; 95 %CI: 0.30–0.47) were significantly less likely to be eligible for LCS than their respective White counterparts.

Under the $USPSTF_{2021}$ guideline, Blacks (AOR: 0.57; 95 %CI: 0.49–0.66) were significantly less likely than Whites to be eligible for LCS. The comparison by sex for LCS eligibility under the $USPSTF_{2021}$ guideline also showed that Black men (AOR: 0.64; 95 %CI: 0.51–0.80) and Black women (AOR: 0.47; 95 %CI: 0.39–0.56) were significantly less likely to be eligible for lung cancer screening than their respective White counterparts.

We developed tailored criteria for Blacks. Under the tailored criteria, Blacks (AOR: 1.00; 95 %CI: 0.88–1.14) had the same odds of eligibility for LCS as Whites. The odds of eligibility for LCS by sex under the tailored criteria did not differ significantly for Black men (AOR: 1.02; 95 %CI: 0.85–1.24) and Black women (AOR: 0.95; 95 %CI: 0.81–1.12) compared to their respective White counterparts.

4. Discussion

We assessed racial/ethnic disparities in LCS eligibility under the 2013 and 2021 USPSTF guidelines using non-institutionalized community data from the 2017–2021 BRFSS survey. The 2021 USPSTF guideline increased the proportion of eligible individuals among both Blacks and whites and slightly reduced racial/ethnic disparities in LCS eligibility compared with the 2013 USPSTF guideline. However, under both guidelines, Black individuals (both men and women) remained less likely to be eligible for LCS than their respective White counterparts.

Several previous studies suggested the need for tailored criteria for Black individuals to increase their odds of eligibility for screening and to bridge the gaps in LCS eligibility between Whites and Blacks [11,17]. However, to our knowledge, none of the previous studies developed tailored criteria for Blacks. The disparities that exist between Blacks and Whites necessitate the development of tailored guidelines for Blacks [11,17]. First, Blacks are less likely to be eligible for LCS than Whites under the existing guidelines [17]. Second, Blacks develop lung cancer at younger ages than their White counterparts [13]. Third, Blacks smoke fewer cigarette than their White counterparts [6,7]. Fourth, Blacks are more likely to be diagnosed with late-stage lung cancer than any other racial/ethnic group, even after the introduction of the USPSTF LCS guideline [4]. Fifth, Blacks are at greater risk of developing lung cancer than Whites [10]. These factors indicate that although Blacks carry the greater burden of lung cancer, they are less likely to benefit from the existing USPSTF LCS guidelines. Tailored criteria for Blacks will bridge the gaps in LCS eligibility between Blacks and White individuals and likely reduce lung cancer mortality among Blacks.

Because Blacks are diagnosed with lung cancer at younger ages and smoke fewer cigarette than their White counterparts [21], it is logical for Blacks to begin LCS at a younger age with lower smoking pack-year histories. The important point here is to identify the appropriate minimum age and smoking pack-years at which Black individuals need to start screening for lung cancer to enable equitable screening rates. To do so, we developed tailored criteria for Blacks by modifying the 2021 USPSTF guideline: we reduced the minimum age from 50 to 43 years and decreased the cumulative number of cigarettes smoked from 20 to 15 pack-years among Blacks. The statistical analysis that we conducted using the tailored criteria showed that there is no significant difference in the odds of eligibility for LCS between Blacks and Whites (including the analysis conducted within strata by sex). Therefore, the tailored criteria eliminate the disparities in LCS eligibility that were seen with the existing guidelines. Moreover, the tailored criteria increased the proportion of eligible Black individuals among ever smoked adults to 26.34 % (9.12 % and 18.49 % under the 2013 and 2021 USPSTF guidelines, respectively). Creating tailored criteria for a specific race/ethnic group is not unprecedented. A tailored criterion was created and recommended for Blacks to undergo colorectal cancer screening at age 45 years, while the recommendation for the remaining racial/ethnic groups was at age 50 years [22,23].

Despite its significant findings, our study is not without limitations. First, the sensitivity and specificity of our tailored criteria were not assessed. Second, BRFSS data were self-reported and may have been subject to recall bias. Third, because the BRFSS data are limited to non-institutionalized individuals with access to a telephone, our findings may not be generalizable to all individuals in the U.S. Fourth, generalizability of the findings could also be limited due to the higher relative number of White participants to Black participants in this study compared to the US population, as well as the exclusion of participants who were missing information on smoking history. Fifth, overdiagnosis and false-positivity could be a concern if the tailored criteria are implemented.

In conclusion, our study found that the 2021 USPSTF guideline increased the proportion of eligible individuals for LCS among both Blacks and Whites compared with the 2013 guideline. Under both

Table 2
Proportion of Eligible individuals among Ever smoked Blacks and Whites.

Guideline	Blacks (%)			Whites (%)		
	Total	Men	Women	Total	Men	Women
$USPSTF_{2013}$	9.12	10.32	8.07	17.26	18.77	15.79
$USPSTF_{2021}$	18.49	19.79	17.36	26.60	26.92	26.27
Tailored*	26.34	27.30	25.49	–	–	–

* Tailored criteria were created only for Blacks.

Table 3
Lung Cancer Screening Eligibility by Race/Ethnicity Using Different Guidelines.

Characteristics	Odds Ratio (95%CI)		
Race/ethnicity	$USPSTF_{2013}$	$USPSTF_{2021}$	Tailored criteria
Both sexes			
Non-Hispanic Whites	Reference	Reference	Reference
Non-Hispanic Blacks	0.46 (0.38–0.54)	0.57 (0.49–0.66)	1.00 (0.88–1.14)
Men			
Non-Hispanic Whites	Reference	Reference	Reference
Non-Hispanic Blacks	0.50 (0.38–0.64)	0.64 (0.51–0.80)	1.02 (0.85–1.24)
Women			
Non-Hispanic Whites	Reference	Reference	Reference
Non-Hispanic Blacks	0.38 (0.30–0.47)	0.47 (0.39–0.56)	0.95 (0.81–1.12)

guidelines, Blacks remained significantly less likely to be eligible for LCS, indicating the need for tailored criteria for Blacks to allow them to benefit equally from LCS. Our study is the first to propose and identify tailored criteria for LCS for Blacks at the age of 43 years with a smoking pack-year history of 15 or more. Our study has shown that the tailored criteria eliminate the disparities between Blacks and Whites in LCS eligibility. Implementing the tailored criteria has the potential to result not only in the elimination of disparities but also in the prevention of many more deaths by detecting lung cancer at early stages. Further studies should test the sensitivity and specificity of the tailored criteria.

CRediT authorship contribution statement

Abdi T. Gudina: Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Formal analysis, Conceptualization. **Charles Kamen:** Writing – review & editing, Formal analysis. **Sara J. Hardy:** Writing – review & editing, Formal analysis. **Lee Kehoe:** Writing – review & editing, Formal analysis. **Eva Culakova:** Writing – review & editing, Formal analysis. **Ana-Paula Cupertino:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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